UNIVERSITY OF MILANO-BICOCCA
Department of Statistics and Quantitative Methods

Course in

Advanced Derivatives

Empirical evaluation of pricing models and no-arbitrage constraints in derivative markets

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Academic Year 2025/2026

Contents

Abstract 2

1 Verification of Merton Bounds, Monotonicity and Convexity Constraints on Mid Prices 3

2 Quadratic Interpolation of Implied Volatility and Black–Scholes Call Valuation 6

3 Monte Carlo Price and 95% Confidence Interval of the Call 8

4 One-Month VSTOXX Estimation Using OTM Options and Comparison with Quoted Value 10

5 EUROSTOXX50 and VSTOXX Returns: Scatterplot, Leverage Effect and Vasicek Estimation 12

6 Estimation and Representation of Normal and Variance Gamma on 3-Month Euribor Returns 15

7 Zero Rate Computation and Discount Curve Construction from EUR OIS 17

8 Pricing of a Callable Credit-Equity Linked Bond 19

Data and Bloomberg Screenshots 23

Abstract

This paper analyzes instruments and data pertaining to derivatives markets with the objective of verifying the consistency of observed prices, reconstructing implied quantities, and applying pricing and calibration techniques. The dataset, extracted from Bloomberg, comprises European Call and Put options on the EUROSTOXX50 with approximately one-month maturity, Euro area interest rate curves (OIS and Euribor), historical time series of the equity index and the VSTOXX volatility index, as well as credit data derived from CDS on the issuer Crédit Agricole.

The first part of this work is devoted to verifying no-arbitrage conditions on option prices. We examine the theoretical Merton bounds, monotonicity and convexity constraints with respect to strike, discussing the role of market frictions and demonstrating how the introduction of tolerances based on the bid–ask spread enables the distinction between purely theoretical violations and genuine arbitrage opportunities.

Subsequently, we reconstruct the implied volatility smile on a single maturity, operating in terms of total volatility and adopting a quadratic interpolation on selected strikes. The volatility thus obtained is employed to price options not directly quoted, comparing the results from the Black–Scholes closed-form formula with Monte Carlo estimates to assess price sensitivity to the volatility input.

A dedicated section addresses the estimation of market volatility: starting from out-of-the-money option prices, we construct an expected volatility measure consistent with the model-free index methodology. We further analyze the relationship between EUROSTOXX50 returns and VSTOXX changes, empirically verifying the leverage effect. To complement the analysis, we calibrate a Vasicek model on log-returns as a benchmark for describing the stochastic dynamics of the series.

Finally, we integrate interest rates and credit risk: through bootstrap of the OIS curve, we obtain risk-free discount factors, while from CDS quotes we extract the issuer's survival probabilities. In this context, we value a callable credit-equity linked bond via Monte Carlo simulation, combining the equity underlying dynamics with the discount curve and credit structure, and discuss the importance of proper management of payoff nonlinearities to achieve a consistent valuation.

1 Verification of Merton Bounds, Monotonicity and Convexity Constraints on Mid Prices

The objective of this section is to verify whether the mid prices of options, computed as the arithmetic mean of bid and ask, satisfy the principal no-arbitrage constraints with respect to strike. The analysis was conducted on prices of European call and put options on the EURO STOXX 50 index with identical maturity. The spot price S_0 equals 5753.9554, corresponding to the index level at the valuation date t_0 (December 11, 2025). The risk-free rate, derived from the EUR OIS curve as the one-month zero rate, equals 1.95634%, while the dividend yield q is set at 2.78%, a value consistent with the T12 Dividend Forecast Yield provided by Bloomberg and with the continuous dividend assumption of the Black–Scholes model.

The Merton bounds were verified through the theoretical lower bounds for calls and puts

$$C(K,T) \geq \max\left(S_0 e^{-qT} - Ke^{-rT}, 0\right) \tag{1}$$

$$P(K,T) \geq \max\left(Ke^{-rT} - S_0 e^{-qT}, 0\right) \tag{2}$$

To incorporate market frictions, the verification was repeated introducing a tolerance threshold defined as the half-spread, i.e., half of the bid–ask spread

$$s(K,T) = \frac{\mathrm{Ask}(K,T) - \mathrm{Bid}(K,T)}{2} \tag{3}$$

Monotonicity was analyzed by verifying the ordering of prices as strike increases. The expected conditions are

$$C(K_{i+1},T) \leq C(K_i,T), \quad i = 1,\ldots,N-1 \tag{4}$$

$$P(K_{i+1},T) \geq P(K_i,T), \quad i = 1,\ldots,N-1 \tag{5}$$

Convexity was tested in discrete form by comparing slopes between adjacent points. Here too, a tolerance was introduced in the slope space, obtained by aggregating the bid–ask semi-differences and normalizing by the strike interval width, thereby making the test robust to microstructure effects. The condition verified for three consecutive strikes (K_{i-1}, K_i, K_{i+1}) is

$$\frac{C_{i+1}^{\mathrm{mid}} - C_i^{\mathrm{mid}}}{K_{i+1} - K_i} \geq \frac{C_i^{\mathrm{mid}} - C_{i-1}^{\mathrm{mid}}}{K_i - K_{i-1}}, \quad \frac{P_{i+1}^{\mathrm{mid}} - P_i^{\mathrm{mid}}}{K_{i+1} - K_i} \geq \frac{P_i^{\mathrm{mid}} - P_{i-1}^{\mathrm{mid}}}{K_i - K_{i-1}} \tag{6}$$

The application of Merton bounds to mid prices reveals few irregularities, concentrated on specific strikes and of overall limited magnitude. On the put side, the bound is violated for strikes 6075, 6100 and 6125 when considering exclusively the mid price. However, upon introducing a tolerance equal to half the bid–ask spread, the bound is satisfied for all strikes in the sample. This indicates that the initial deviations are compatible with market frictions and do not signal a structural inconsistency in prices.

Regarding monotonicity with respect to strike, the price curve behaves regularly. Calls decrease as K increases and puts increase, without local inversions. This confirms that

the observed surface is correctly aligned along the strike dimension and does not exhibit anomalies typical of noisy data or quotation errors.

On the convexity front, the test conducted on mid values reveals some irregularities. For calls, the constraint is not satisfied at strike 5475, while for puts the violations are concentrated at strikes 6000 and 6075. Again, upon introducing the spread-related tolerance, the deviations are absorbed and the surface proves compatible with theoretical convexity.

The discrepancies that emerge when working solely on mid prices should be interpreted as quotation deviations rather than effectively exploitable arbitrage opportunities. Their disappearance after introducing the tolerance suggests that any strategy aimed at exploiting them would be neutralized by transaction costs and bid–ask uncertainty. The analysis further highlights sensitivity to parameter choice, particularly the dividend yield. Adopting an observed market-consistent value renders the bounds more stringent and may reveal small deviations on the mid, which tend to appear precisely where spreads are widest. This confirms the central role of liquidity in the empirical interpretation of no-arbitrage constraints. Once frictions are properly incorporated, the overall price structure proves consistent with theoretical conditions.

The plot of call and put mid prices as a function of strike provides an immediate qualitative check on surface plausibility. In a regular market, calls decrease with K because higher strikes reduce the probability of finishing in-the-money, while puts increase with K since higher strikes raise the probability of a positive payoff at expiry. The region where the two point sets tend to converge is that proximate to the ATM strike value, confirming that the data are well-aligned and free of macroscopic anomalies.

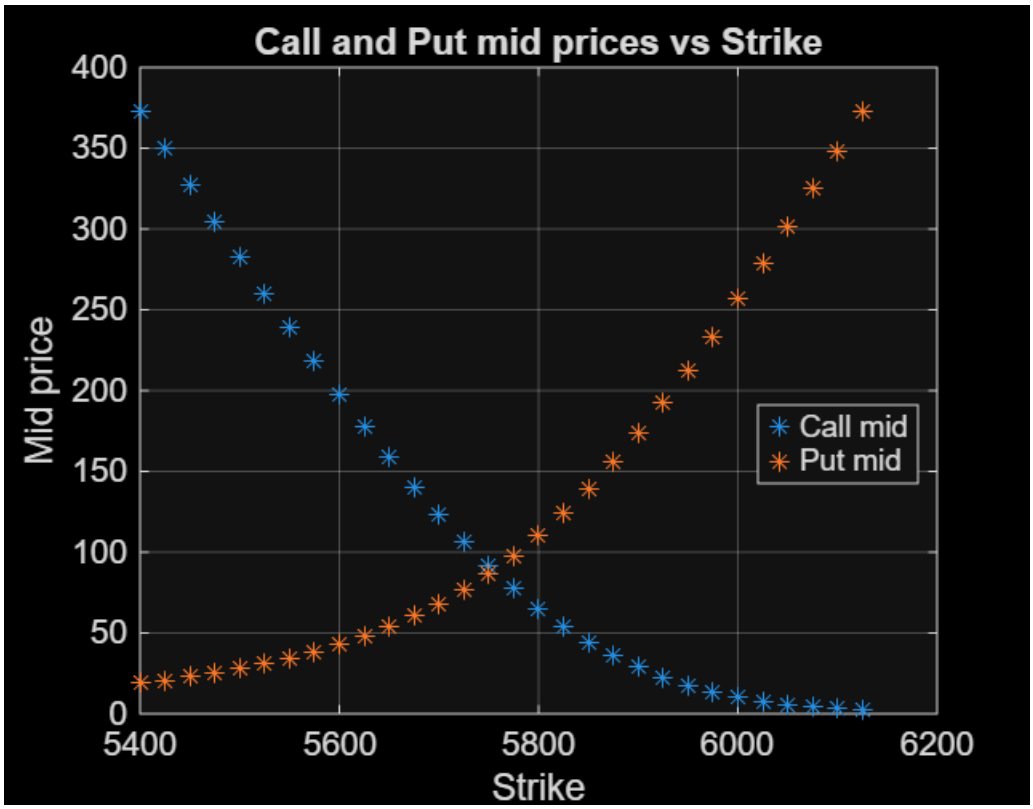

Figure 1: Mid prices of Call and Put options as a function of strike.

2 Quadratic Interpolation of Implied Volatility and Black–Scholes Call Valuation

The objective of this section is to estimate, in a market-consistent manner, the implied volatility associated with a European call with non-quoted strike $K^* = 5753$ and compute its theoretical price. The maturity of the observed options is $T = 36/365$, corresponding to the interval between December 11 and January 16.

The adopted procedure locally reconstructs the volatility smile from the implied volatilities available from Bloomberg, then applies the Black–Scholes formula with continuous dividends. The logic is entirely market-oriented: we start from the volatilities of quoted options and reconstruct the smile to obtain a consistent estimate at K^*.

The estimation was conducted on a selected sample, composed exclusively of out-of-the-money (OTM) Call options. This methodological choice is motivated by the greater liquidity of such quotes, which reduces the informational noise often present in in-the-money options and permits a more faithful reconstruction of the smile.

The interpolation was not applied directly to raw volatilities, but rather to total volatility, defined as

$$v(K) = \sigma_{\text{imp}}(K)\sqrt{T} \tag{7}$$

The functional relationship between $v(K)$ and strike K was modeled via quadratic regression

$$v(K) = \beta_0 + \beta_1 K + \beta_2 K^2 \tag{8}$$

Analysis of the estimated coefficients reveals a regular pattern in the volatility term structure. The constant term captures the average level of implied volatility. The linear term, which is negative, reflects the skew slope typical of equity markets, where OTM Puts tend to have relatively higher prices compared to OTM Calls. The quadratic term captures the moderate curvature (smile) observed in the domain of analyzed options.

From a methodological standpoint, the adopted approach deviates slightly from the theoretical Shimko model in tail management, as we did not impose the condition that $v(K)$ becomes constant outside the observed strikes. This choice, dictated by graphical representation requirements, does not compromise the validity of the point estimate, as the value at K^* depends exclusively on local interpolation over available market data.

Evaluating the interpolated function at the target strike $K^* = 5753$ yields an implied volatility of

$$\sigma_{\text{impl}}(K^*) \approx 13.36\% \tag{9}$$

This value is consistent with the volatilities observed on adjacent quoted strikes, confirming the goodness of the quadratic fit to the market structure.

The Call option price was computed using the Black & Scholes analytical formula, with the interpolated volatility σ_{impl} as input, the risk-free rate r derived from the OIS curve, and the continuous dividend yield q observed on Bloomberg. The formula employed is

$$C(S_0, K^*, T) = S_0 e^{-qT} N(d_1) - K^* e^{-rT} N(d_2) \tag{10}$$

where d_1 and d_2 are computed using $\sigma = 0.1336$.

The resulting Call option price is

$$C_{BS} \approx 94.22 \tag{11}$$

The obtained value is fully in line with market quotations for near-the-money options with analogous maturity, confirming the economic plausibility of the result. The quadratic interpolation on OTM Calls alone provided a numerically stable volatility estimate, enabling the pricing of an illiquid strike in a manner consistent with no-arbitrage conditions and the prevailing market volatility structure.

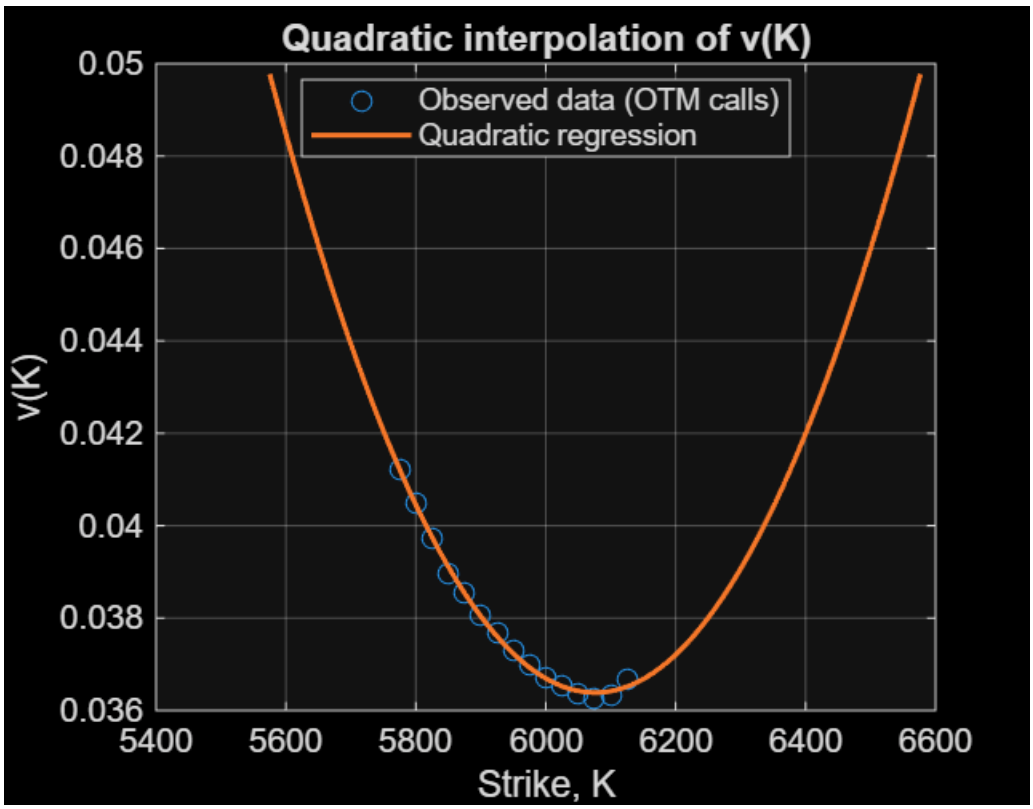

Figure 2: Observed implied volatilities as a function of strike and quadratic curve interpolated on $v(K)$.

3 Monte Carlo Price and 95% Confidence Interval of the Call

The objective of this section is to price a European call via Monte Carlo simulation with 20,000 paths, given strike $K = 5740$ and volatility $\sigma = 0.07$, and to compare the result with the price obtained from the Black–Scholes formula using the volatility interpolated in the previous section.

The valuation was conducted at date t_0 = December 11, 2025, with underlying S_0 = 5753.9554 and expiry January 16, 2026, corresponding to $T = 36/365$ years.

The risk-free rate was obtained via bootstrap of the one-month zero rate from the EUR OIS curve. Starting from the bid and ask quotes of the one-month OIS, the mid rate was computed, from which the discount factor and subsequently the zero rate in continuous compounding were derived. This procedure is consistent with market practice for pricing short-dated instruments and ensures consistency between the rate structure employed and the option's time horizon.

The dividend yield employed is $q = 2.78\%$, corresponding to the T12 DF Yield of the EURO STOXX 50 indicated by Bloomberg at the valuation date, consistently with the continuous dividend yield assumption required by the Black–Scholes model. The use of consistent market parameters for risk-free rate and dividend yield contributes to rendering the results economically plausible and comparable across different valuation methods.

The call price was computed via Monte Carlo simulation with 20,000 paths and volatility $\sigma = 0.07$. The resulting price is approximately 55 (54.9964), with a 95% confidence interval of [53.9, 56.1]. The price computed with the Black–Scholes formula, using the same volatility, equals approximately 55.0 (54.9725) and falls fully within the Monte Carlo confidence interval. This result confirms the correct implementation of the simulation and the consistency between the two pricing methods when the underlying assumptions, particularly volatility, are identical.

The analysis was extended by comparing prices obtained with both methods using also the implied volatility interpolated in the previous section. The interest was to verify whether the consistency observed between Black–Scholes and Monte Carlo would be maintained with a market-derived volatility input.

Using the implied volatility interpolated from OTM calls, the call price equals approximately 101.6 (101.6212) with Black–Scholes and 101.7 (101.7026) with Monte Carlo. Here too, the two estimates are extremely close, indicating that the observed price differences are attributable exclusively to the different volatility employed and not to the pricing method used.

The graphical comparison clearly highlights this result. At equal volatility, Black–Scholes and Monte Carlo produce nearly identical values, while the transition from test volatility to market-implied volatility entails a significant increase in option price. This underscores the strong sensitivity of call value to volatility and confirms that, in a risk-neutral context, the choice of volatility input is the primary price determinant, more so than the valuation technique adopted.

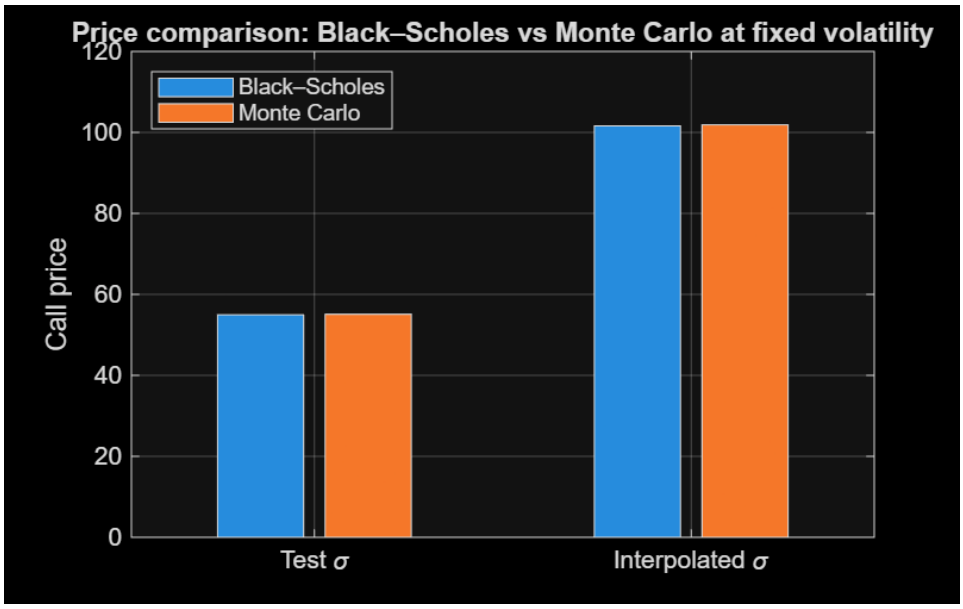

Figure 3: Comparison between Monte Carlo and Black–Scholes prices with test volatility and interpolated volatility.

4 One-Month VSTOXX Estimation Using OTM Options and Comparison with Quoted Value

The objective of this section is to construct a didactic estimate of the VSTOXX at approximately one month using a single maturity and only out-of-the-money (OTM) options.

The parameters considered are $t_0 =$ December 11, 2025, expiry January 16, 2026, and time to maturity computed using the ACT/360 convention, yielding $T = 0.10$. The EUROSTOXX50 level at t_0 is $S_0 = 5753.9554$. For each strike, the option mid price $O(K)$ is constructed as the average of bid and ask, selecting only OTM options and separating puts with $K \leq S_0$ from calls with $K > S_0$.

On this grid of OTM strikes, we apply the one-maturity formula, a discrete sum approximating the implied variance integral

$$\mathrm{VIX}_T \approx 100\sqrt{\frac{2}{T}\sum_{i=1}^{n}\frac{\Delta K_i}{K_i^2}O(K_i)} \tag{12}$$

where K_i are the sorted OTM strikes, ΔK_i the differences between consecutive strikes, and $O(K_i)$ the mid prices of OTM options. We are not estimating historical volatility from past returns, but rather implied volatility, i.e., the volatility that the market is incorporating into option prices for horizon T. The result is annualized and reported in percentage terms via the factor 100.

The formula employed is a simplified version of the complete construction. The implied variance for maturity T is typically written in the form

$$\sigma_T^2 = \frac{2}{T}\sum_i \frac{\Delta K_i}{K_i^2}e^{rT}O(K_i) - \frac{1}{T}\left(\frac{F}{K_0} - 1\right)^2 \tag{13}$$

$$\text{with VIX} = 100\sqrt{\sigma_T^2} \tag{14}$$

Relative to the complete formula, we observe that:

the OTM selection and reference point near ATM are linked to the forward F and strike K_0 close to F, not to spot S_0;

the factor e^{rT} appears, incorporating the risk-free rate carry effect;

a correction term $\left(\frac{F}{K_0} - 1\right)^2$ appears, derived from the theoretical variance replication.

The one-maturity formula used is equivalent to assuming these adjustments are negligible (e.g., $r \approx 0$ and $F/K_0 \approx 1$), focusing on the dominant component given by the sum of OTM prices weighted by $\Delta K_i/K_i^2$.

Application of the procedure yields an implied volatility index estimate of 12.6873. Comparison with the market-observed value on the same date, equal to 15.0399, reveals a negative deviation of approximately -2.3526, i.e., a percentage change greater than 15% relative to the observed market price. The reconstructed implied volatility is more contained than that incorporated in the quoted index, signaling that the obtained measure captures a

lower level of expected risk than that reflected by the market.

The discrepancy between estimated and observed values is interpretable as a natural consequence of numerical approximation and the operational choices underlying index construction. The estimate is particularly sensitive to the representation of the price structure in the tails. When strike coverage is limited and does not extend sufficiently toward very low or very high levels, part of the variance contribution is not fully incorporated, with a typical underestimation effect. The market measure, conversely, is constructed to be as robust as possible and adherent to a standardized definition of the reference horizon, while a didactic reconstruction tends to prioritize simplicity and replicability over completeness of the steps present in official specifications.

5 EUROSTOXX50 and VSTOXX Returns: Scatterplot, Leverage Effect and Vasicek Estimation

This section jointly analyzes the dynamics of EUROSTOXX50 and VSTOXX returns with a dual objective. On one hand, we aim to empirically verify the presence of the leverage effect through the correlation between log-returns; on the other, we study the temporal dynamics of returns through an OLS calibration of the Vasicek model applied directly to the log-return series.

Leverage Effect: Correlation Between Market Returns and Volatility

Starting from prices aligned on the same dates, log-returns of EUROSTOXX50 and VSTOXX were constructed. The dependence between the two series was analyzed via scatterplot and linear correlation coefficient. The estimated correlation value is $\rho \approx -0.79$ with an extremely small p-value ($\sim 10^{-107}$), indicating that the observed negative relationship is statistically highly significant.

This result is perfectly consistent with the leverage effect documented in the literature. When the equity market registers negative returns, implied volatility tends to increase markedly. From an economic perspective, this phenomenon reflects both the increase in firms' financial leverage during downturns and the rise in demand for protection via options (particularly puts), which translates into a VSTOXX increase.

The log-return scatterplot clearly shows this structure. The points are distributed along a negatively inclined cloud, with positive volatility changes associated with negative equity returns and vice versa. The presence of some outliers is natural in financial returns but does not alter the underlying signal, which is clear and robust.

OLS Calibration of the Vasicek Model on Log-Returns

Log-returns were modeled as a discrete Vasicek-type process, exploiting the Euler discretization of the continuous model. The linear regression

$$r_{i+1} = a\, r_i + b + \varepsilon_i \tag{15}$$

allows derivation of the model's structural parameters

$$\lambda = \frac{1-a}{\Delta t}, \quad \mu = \frac{b}{1-a}, \quad \sigma^2 = \frac{\mathrm{Var}(\hat{\varepsilon})}{\Delta t} \tag{16}$$

For EUROSTOXX50 log-returns we obtain $a \approx -0.026$, $\lambda \approx 1.03$, $\mu \approx 0.00048$ and $\sigma \approx 0.0095$.

The coefficient a, which is close to zero and slightly negative, suggests at most negligible serial dependence in daily returns. Consistently with this finding, the OLS scatter plots (Figure 5) display a nearly horizontal fitted line and a highly dispersed point cloud, indicating that the linear Vasicek-type specification applied to returns carries virtually no explanatory power. Even without reporting it explicitly, the corresponding R^2 is likely to be indistinguishable from zero: r_i contains essentially no useful information for predicting r_{i+1} in the

conditional mean. Under this parameterization, the transformation $\lambda = (1 - a)/\Delta t$ yields values of λ close to unity (with $\Delta t = 1$), but this result should be interpreted as a mechanical consequence of $a \approx 0$ rather than evidence of economically meaningful mean reversion. Within the Ornstein–Uhlenbeck framework, such a value would imply an extremely short half-life on the order of a single trading day, suggesting that any serial dependence dissipates almost instantaneously and is therefore not economically identifiable. The estimate of μ is close to zero, as expected at daily frequency, while σ is of a magnitude consistent with a broad, liquid equity index.

For the log-returns of the VSTOXX, the estimates are $a \approx -0.0024$, $\lambda \approx 1.00$, $\mu \approx 0.00026$, and $\sigma \approx 0.068$. Here again, a is effectively indistinguishable from zero and the OLS fit is extremely poor, reinforcing the conclusion that a Vasicek specification applied to returns fails to capture the relevant dynamic structure of the series. Notably, the estimated volatility is an order of magnitude larger than that obtained for the EUROSTOXX 50, which is consistent with the nature of the VSTOXX as an index of implied volatility expectations and its well-documented sensitivity to sharp movements and extreme observations.

Taken together, the two analyses yield a coherent picture. At the cross-sectional level, the negative relationship between equity returns and changes in implied volatility is strong and well-identified, providing clear evidence of a leverage effect. From a time-series perspective, however, the data offer little support for conditional mean predictability through a Vasicek-type regression on returns. The dominant empirical features are instead the high variability of the VSTOXX and the heavy-tailed, heteroskedastic behavior characteristic of financial returns, properties that are more appropriately captured by time-varying volatility specifications, such as GARCH-class models.

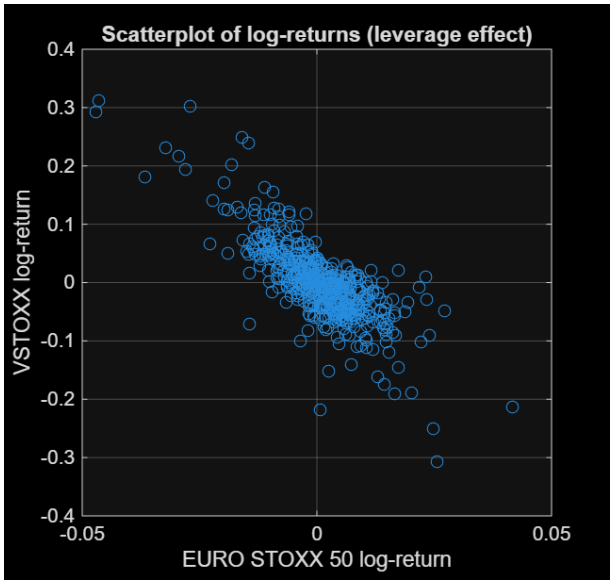

Figure 4: Scatterplot of EUROSTOXX50 and VSTOXX log-returns.

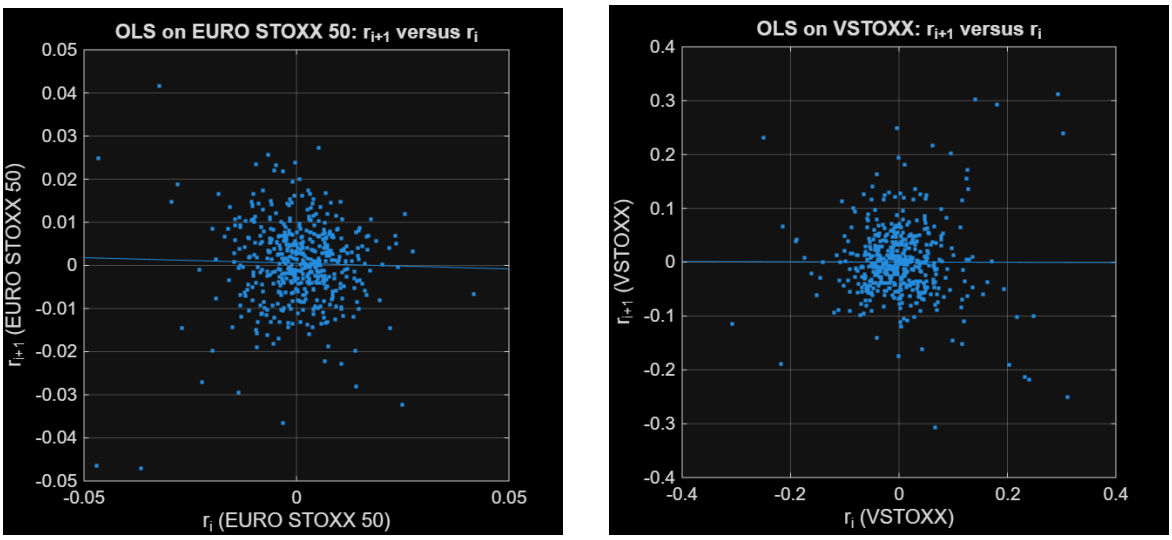

Figure 5: OLS regression on log-returns: EUROSTOXX50 (left) and VSTOXX (right).

6 Estimation and Representation of Normal and Variance Gamma on 3-Month Euribor Returns

The objective of this section is to analyze the dynamics of 3-month EURIBOR returns starting from the available historical series. The purpose is twofold. On one hand, we construct the empirical distribution of returns by applying two different transformations (simple returns and log-returns); on the other, we estimate a Normal distribution and a Variance Gamma distribution on the empirical distribution, comparing their parameters and goodness-of-fit via graphical analysis and statistical tests.

Missing values were handled with forward-fill ("previous"), a choice consistent with maintaining continuity in the rate series. On this basis, two transformations of daily dynamics were computed

1. Simple returns: $\Delta y_t = y_t - y_{t-1}$
2. Log-returns: $\Delta \log(y_t) = \log(y_t) - \log(y_{t-1})$

For each return series, a comparison was conducted between a Normal distribution, estimated via maximum likelihood, and a Variance Gamma (VG) distribution, estimated by numerically maximizing the log-likelihood with respect to the parameter vector $[m_0, \text{drift}, \sigma, \text{shape}]$.

For the Normal distribution, in the case of simple returns the estimation yields $\mu = -0.0037012$ and $\sigma = 0.0178578$. For log-returns, the estimated parameters are $\mu = -0.00126255$ and $\sigma = 0.00668799$.

For the Variance Gamma, in the case of simple returns we obtain $m_0 = 0.00150071$, drift $= -0.00157234$, $\sigma = 0.0095931$ and shape $= 3.30836$. For log-returns we obtain $m_0 = -0.000485246$, drift $= -0.000387663$, $\sigma = 0.00469322$ and shape $= 2.0051$.

The shape value exceeding 3 for simple returns and approximately 2 for log-returns suggests a form that deviates significantly from the Gaussian, compatible with a more flexible distribution capable of reproducing leptokurtosis and heavier tails.

To support the distributional analysis, two complementary normality tests were conducted, each designed to capture different aspects of the empirical distribution. The Kolmogorov–Smirnov test verifies whether the empirical distribution of the data is compatible with a theoretical reference distribution, in this case the Normal. It does so by comparing, point by point, the empirical cumulative distribution function with the theoretical one, seeking the maximum discrepancy between the two. It is a test particularly sensitive to the body of the distribution, i.e., the central region where most observations are concentrated. In our case, the test does not reject the null hypothesis at the 5% significance level neither for simple returns (p = 0.09987) nor for log-returns (p = 0.1462), suggesting that, at least in the central part of the distribution, the Normal is not an entirely unacceptable approximation.

The Jarque–Bera test instead operates on a different plane: rather than comparing the distributions in their entirety, it focuses exclusively on sample skewness and kurtosis, verifying whether these deviate in a statistically significant way from the values expected under the Gaussian hypothesis. In this sense it is a targeted test, which interrogates the shape of the distribution in its tail and symmetry properties. The results are clear: the test rejects

normality in both cases, indicating that sample skewness and kurtosis are incompatible with the Gaussian hypothesis. The two tests, far from being contradictory, actually tell the same story from different angles. The empirical distribution resembles the Normal in its average behavior, but deviates markedly in the tails and in the overall shape, exactly where financial risk manifests with greater intensity.

This reading is fully confirmed by the graphical analysis. The QQ-plots reported in Figure 6 show a satisfactory alignment between the empirical quantiles and the theoretical normal quantiles in the central region, but an evident deviation in the tails, both in the lower and in the upper one. Extreme observations deviate systematically from the theoretical line, a clear signal of heavier tails than Gaussian ones. Similarly, the density plots in Figure 7 highlight how the empirical distribution exhibits a more pronounced central peak than the Normal, flanked by thicker tails: the characteristic signature of leptokurtosis. In both transformations considered, the Variance Gamma distribution overlaps the empirical distribution in a visibly more faithful way than the Normal, adapting more precisely both to the central peak and to the tail regions, which are precisely the most critical areas for risk assessment and for the modeling of extreme events.

The Variance Gamma, thanks to its structural flexibility, therefore confirms itself as a more adequate distributional specification to capture the empirical properties of 3-month EURIBOR returns, and more generally of high-frequency financial returns, characterized by asymmetry, excess kurtosis, and non-Gaussian behavior in the tails.

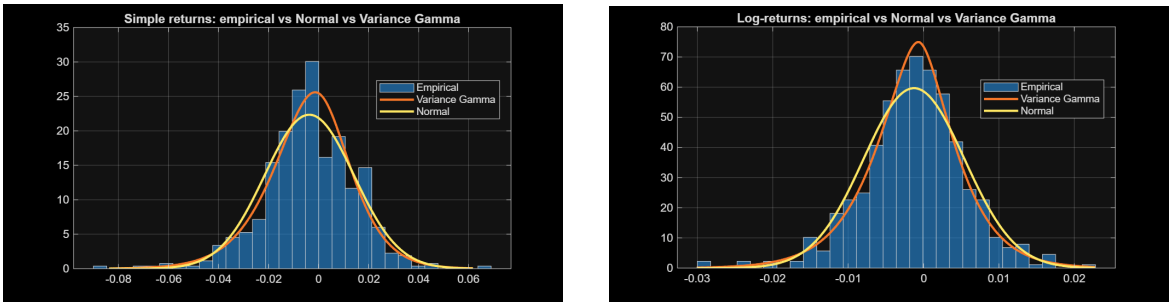

Figure 6: QQ-plot: simple returns (left) and log-returns (right).

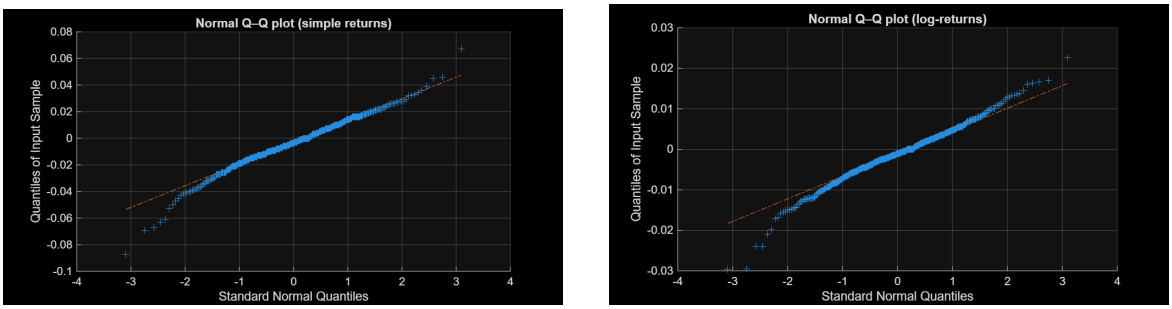

Figure 7: Empirical distribution and fit: simple returns (left) and log-returns (right).

7 Zero Rate Computation and Discount Curve Construction from EUR OIS

The objective of this section is to construct the EUR OIS curve via bootstrap, starting from mid quotes (bid/ask average) and recursively deriving discount factors $P(0,T)$, from which zero rates in continuous compounding are obtained.

To derive the formulas for computing discount factors, the no-arbitrage condition is imposed on the quoted OIS instruments, assuming mid quotes (bid/ask average) as an estimate of the market fair rate. From the cash-flow structure of the contracts, for short maturities the discount factor is obtained through the single-cash-flow relation, while for longer maturities one recursively derives $P(0,T_n)$ by imposing the OIS swap par condition. This leads to the following formulas for the different maturities.

For short maturities ($T < 1$ year), as we said before, the instrument is treated as a contract with a single cash flow at maturity

$$P(0,T) = \frac{1}{1 + R^{\text{mid}}(T)\,\tau(0,T)} \tag{17}$$

For longer maturities ($T \geq 1$ year), the OIS swap parity condition is applied

$$P(0,T_n) = \frac{1 - R^{\text{mid}}(T_n)\sum_{i=1}^{n-1}\tau(T_{i-1},T_i)\,P(0,T_i)}{1 + R^{\text{mid}}(T_n)\,\tau(T_{n-1},T_n)} \tag{18}$$

Once the curve $P(0,T)$ is obtained, zero rates are converted to continuous compounding via

$$z(T) = -\ln(P(0,T))/T \tag{19}$$

It is important to clarify that, while the bootstrap structure is correct, the implementation introduces some simplifications. Maturity and day count management is made operational through fixed conversions (weeks and months transformed into days with constant rules and years assumed to be 360 days). Beyond one year, an annual-frequency fixed leg is assumed with treatment of any residual periods via year fractions. For valuation of long-dated swaps requiring discount factors on intermediate non-quoted dates, log-linear interpolation of discount factors is adopted (linear interpolation on $\log P(0,T)$), which tends to preserve positivity and monotonicity of the curve.

The bootstrap procedure on the EUR OIS curve enables obtaining a discount curve and corresponding zero rates in continuous compounding in an internally consistent manner directly usable for pricing. Any differences relative to Bloomberg values are interpretable as effects of simplified implementation conventions, but do not compromise the economic validity of the approach.

Table 1: Bootstrap results on the EUR OIS curve.

Tenor	Discount Factor	Zero Rate	Zero Rate (%)
1W	0.999625	0.019306	1.9306
2W	0.999250	0.019303	1.9303
1M	0.998393	0.019294	1.9294
2M	0.996789	0.019299	1.9299
3M	0.995188	0.019296	1.9296
6M	0.990421	0.019250	1.9250
1Y	0.980895	0.019290	1.9290
2Y	0.960477	0.020163	2.0163
5Y	0.891045	0.023072	2.3072
10Y	0.767045	0.026521	2.6521
20Y	0.654529	0.021192	2.1192
30Y	0.613884	0.016265	1.6265
50Y	0.588458	0.010605	1.0605

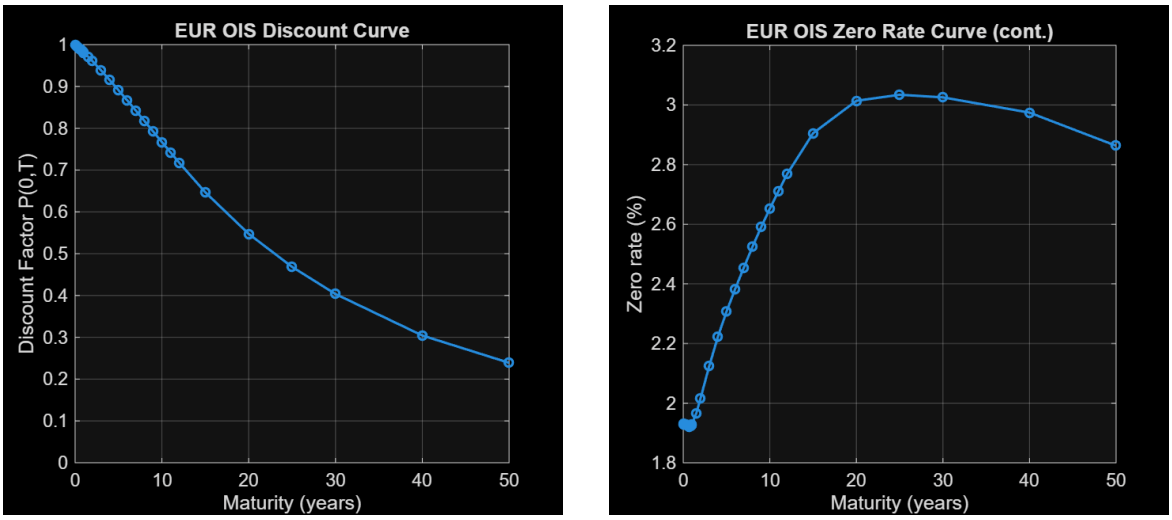

Figure 8: EUR OIS discount curve (left) and zero rate curve (right).

8 Pricing of a Callable Credit-Equity Linked Bond

The objective of this section is to estimate in a risk-neutral framework the value of a bond issued by Crédit Agricole with notional $N = 100$, 4-year maturity and semi-annual coupons. The payment profile of the security combines cash flows dependent on the performance of the EUROSTOXX50 index and the issuer's credit risk.

The temporal structure of the contract provides for semi-annual observation and payment dates. At each date t_i, a coupon equal to $N \cdot C\%$ is paid if the ratio between the underlying level at that date and the initial level exceeds the threshold

$$\frac{S_{t_i}}{S_{t_0}} > 0.95 \tag{20}$$

The issuer may also call the bond early, paying the notional N, if

$$\frac{S_{t_i}}{S_{t_0}} > 1.15 \tag{21}$$

In the event of early redemption, subsequent cash flows are cancelled and the bond value incorporates only cash flows up to the first exercise date.

Market inputs not derivable in a structured manner from the files were entered manually. The initial underlying level $S(t_0)$ was read from a Bloomberg screen (GP Historical of EUROSTOXX50), while the recovery rate $R = 40\%$ was read from the Bloomberg screen for the issuer's CDS (CDSV). This choice ensures traceability and replicability of the analysis.

For the risk-free component, cash flows are discounted using the EUR OIS curve. Starting from zero rates, discount factors $DF(0, T)$ are constructed and values at semi-annual maturities are derived via interpolation on log-discount factors to preserve numerical stability.

The equity component is simulated via a Geometric Brownian Motion, the standard model used to describe the stochastic evolution of a stock price over time. The simulation is performed on a daily time grid, assuming 252 trading days per year, and generates 20,000 underlying paths, a sufficiently large number to ensure stability and accuracy in the Monte Carlo estimates. The annual volatility σ is computed from the stock's historical returns and then annualized according to the market convention. The drift, which represents the expected growth rate of the underlying under the risk-neutral measure, incorporates the risk-free rate extracted from the OIS curve, reduced by the dividend yield q to account for the cash flows distributed over the product's lifetime.

Credit risk is introduced via a survival curve $S(t) = P(\tau > t)$, with cumulative default probability $Q(t) = 1 - S(t)$. The curve was constructed in two variants. The first derives directly from default probabilities provided as input from Bloomberg. The second is obtained via bootstrap from CDS spreads, determining an implied hazard rate such that the CDS value is zero, balancing premium and protection legs under the OIS discount curve with a consistent recovery rate.

The equity component is simulated under the risk-neutral measure via a daily discretization of the Geometric Brownian Motion, according to the scheme:

$$S_{t+\Delta t}^{(j)} = S_t^{(j)} \cdot \exp\left[\left(r - q - \frac{\sigma^2}{2}\right)\Delta t + \sigma\sqrt{\Delta t}\, Z_t^{(j)}\right] \tag{22}$$

where r denotes the OIS risk-free rate, q the dividend yield, σ the annualized historical volatility, and $Z_t^{(j)} \sim \mathcal{N}(0,1)$ are independent standard normal draws. The simulation is run on a daily grid ($\Delta t = 1/252$) over $N = 20{,}000$ paths, allowing precise extraction of the underlying level at each semi-annual observation date $t_1, \ldots, t_8$.

Management of payoff conditionality was addressed with two methodologies. In the first approach, at dates t_i the underlying is replaced with the cross-sectional mean of the simulated paths, and the trigger condition is evaluated on this averaged quantity, yielding deterministic active or inactive cash flows per date:

$$P^{\text{avg}} = \sum_{i=1}^{i^*} C_i\left(\frac{1}{N}\sum_{j=1}^{N} S_{t_i}^{(j)}\right) \cdot B(0,t_i) \cdot Q(t_i) \tag{23}$$

where $B(0,t_i)$ is the OIS discount factor and $Q(t_i)$ the issuer survival probability at date t_i. In the second approach (pathwise), the condition is evaluated for each path j and each date t_i, generating a path-specific set of cash flows and terminating all subsequent cash flows after the first potential early redemption date:

$$P^{\text{path}} = \frac{1}{N}\sum_{j=1}^{N}\sum_{i=1}^{i^*(j)} C_i\left(S_{t_i}^{(j)}\right) \cdot B(0,t_i) \cdot Q(t_i) \tag{24}$$

The price estimate is given by the mean of present values across individual paths, and the availability of the present value distribution enables estimation of a 95% Monte Carlo confidence interval via the standard error of the mean.

It is important to emphasize that, for nonlinear payoffs constructed with indicator functions, the path-average approach is not in general equivalent to the mean of pathwise valuations. The two formulas differ in the order of operations: in the first case the mean is taken *inside* the nonlinear function $C_i(\cdot)$, whereas in the second it is taken *outside*. By Jensen's inequality, $C_i(\mathbb{E}[S]) \neq \mathbb{E}[C_i(S)]$ for nonlinear C_i, and this discrepancy is the source of the large price gap documented below.

The results show negligible impact from the choice of credit curve (from file or bootstrap), but a very marked impact from the methodological choice. The price obtained with the path-average method equals 120.65 using the survival curve from file and 120.69 with bootstrap. Pathwise valuation produces a price of 102.65 in the non-bootstrap case, with a 95% Monte Carlo confidence interval of [102.54, 102.76], and 102.67 in the bootstrap case, with an interval of [102.56, 102.78].

The deviation between the two credit curves is on the order of a few cents in both approaches, indicating that the implied survival profiles are substantially aligned at the relevant maturities. The approximately 18-point difference between the path-average and pathwise methods is consistent with the strong nonlinearity induced by the activation thresholds and the early redemption clause: the path-average approach artificially compresses the dispersion of the simulated underlying, altering the implied frequency of coupon activations

and calls. The narrowness and overlap of the pathwise confidence intervals confirm the numerical stability of the simulation with 20,000 paths.

The analysis reveals high robustness with respect to the default curve construction mode (direct or bootstrap) and, at the same time, shows that proper pathwise management of payoff conditionality is essential to obtain an economically representative valuation of a security with nonlinear threshold-based payoffs. Values between 102.65 and 102.67 represent the standard Monte Carlo estimate of the payoff, while values around 120.65 and 120.69 should be interpreted as the result of a deterministic approximation based on averages.

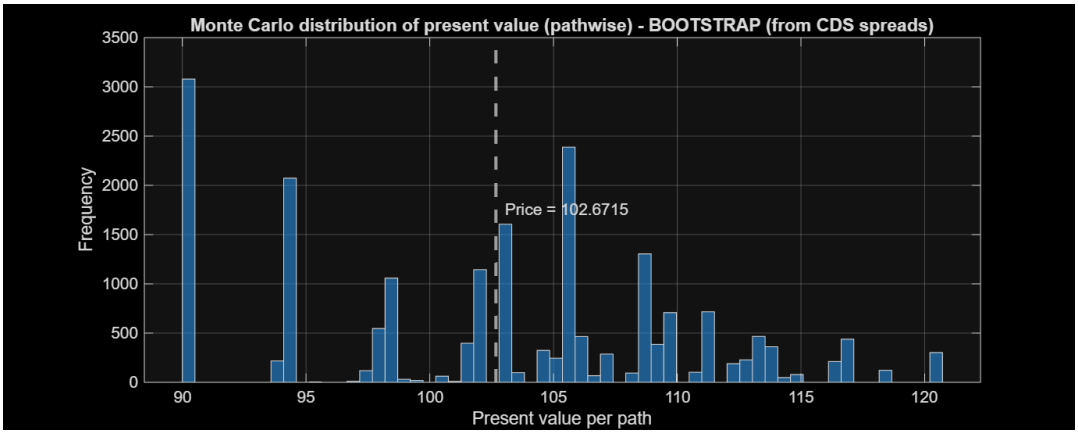

Figure 9: Monte Carlo distribution of present value: with bootstrap.

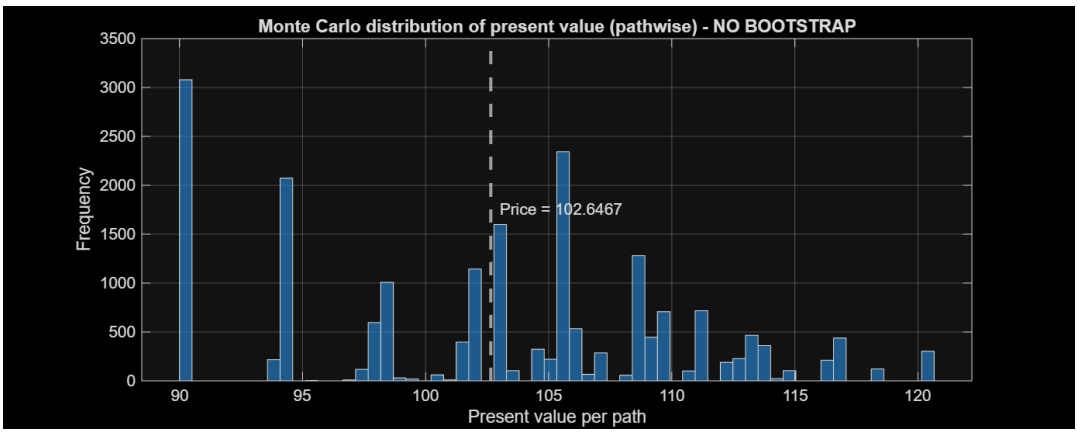

Figure 10: Monte Carlo distribution of present value: without bootstrap.

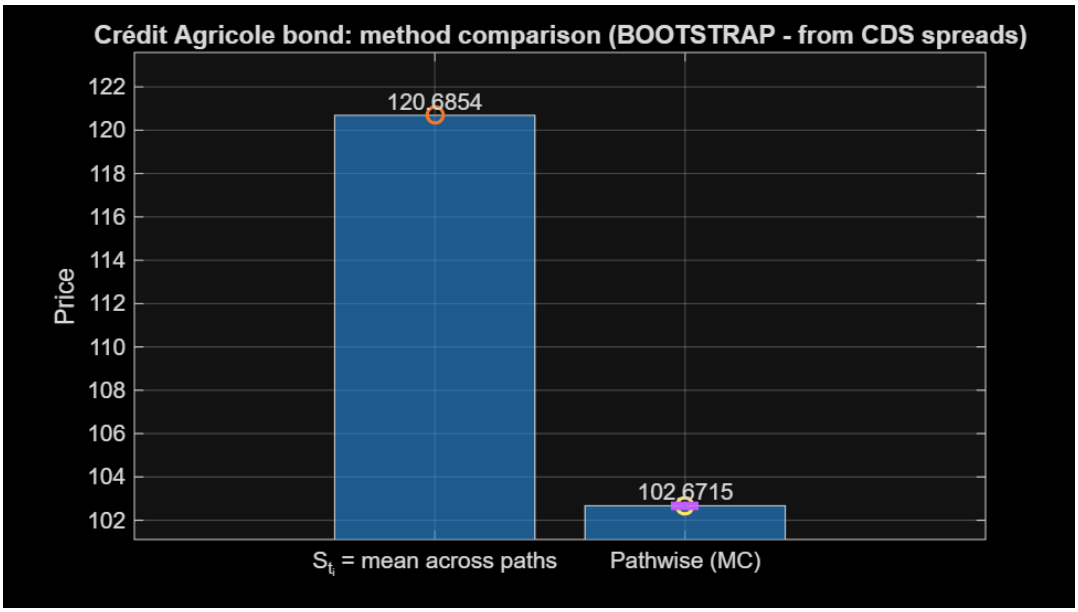

Figure 11: Comparison between the two valuation methods: bootstrap case.

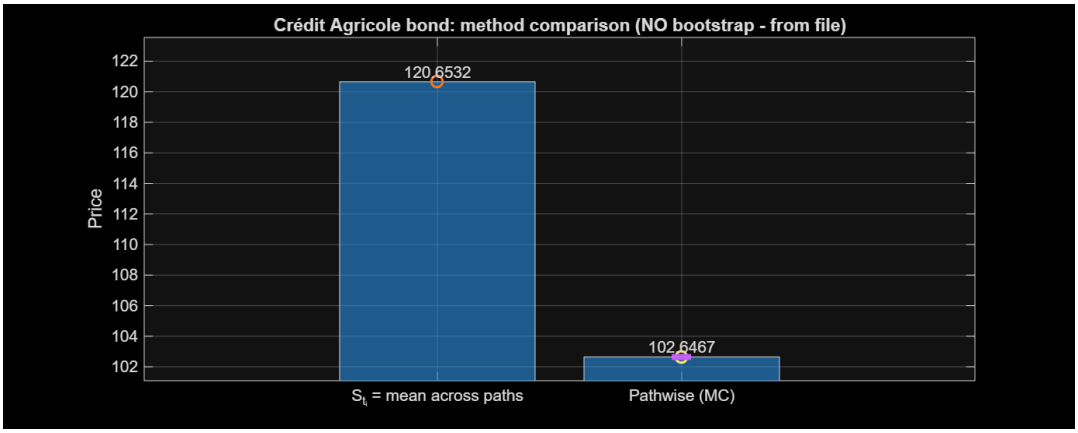

Figure 12: Comparison between the two valuation methods: non-bootstrap case.

Data and Bloomberg Screenshots

Below are the Bloomberg screenshots of the data used for the assignment. The images document OIS swap rate quotes, zero rates, Call and Put option prices, index historical series, and credit information pertaining to the issuer.

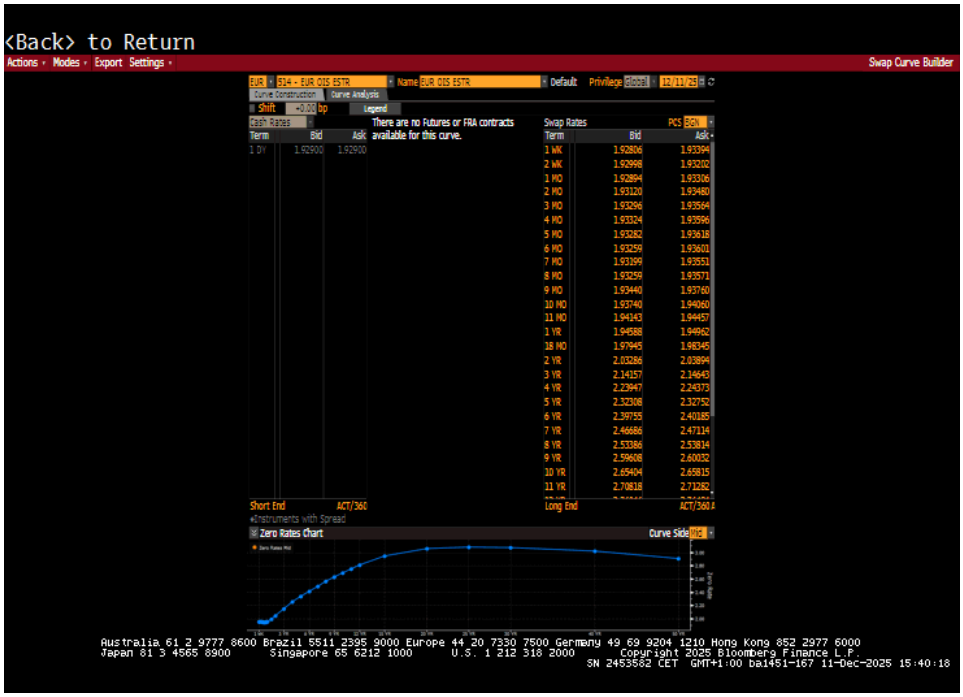

Figure 13: Bloomberg screenshot: OIS swap rate quotes.

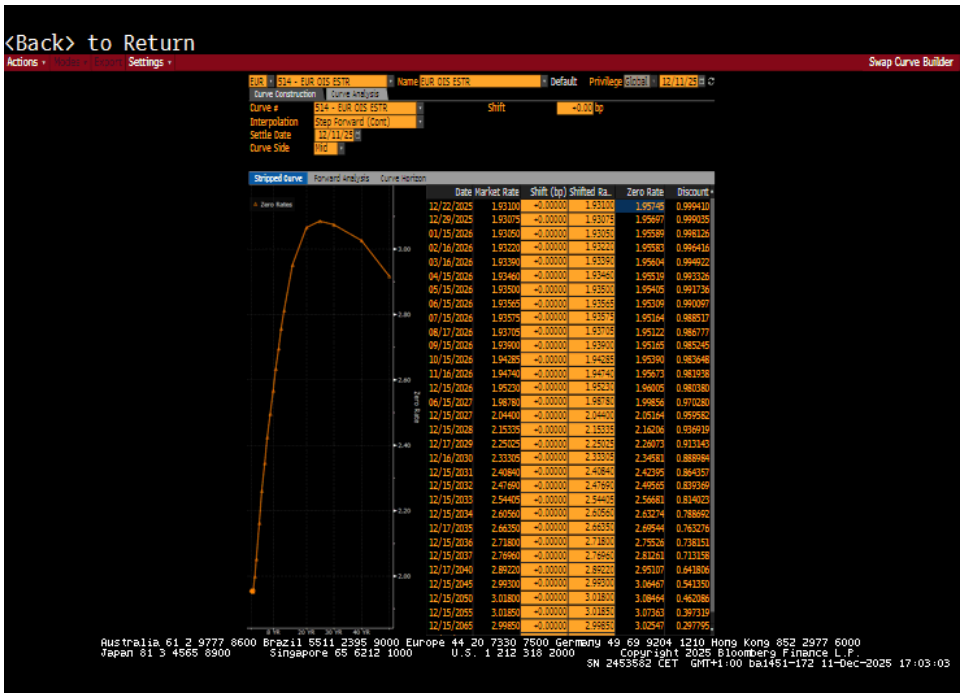

Figure 14: Bloomberg screenshot: OIS zero rates.

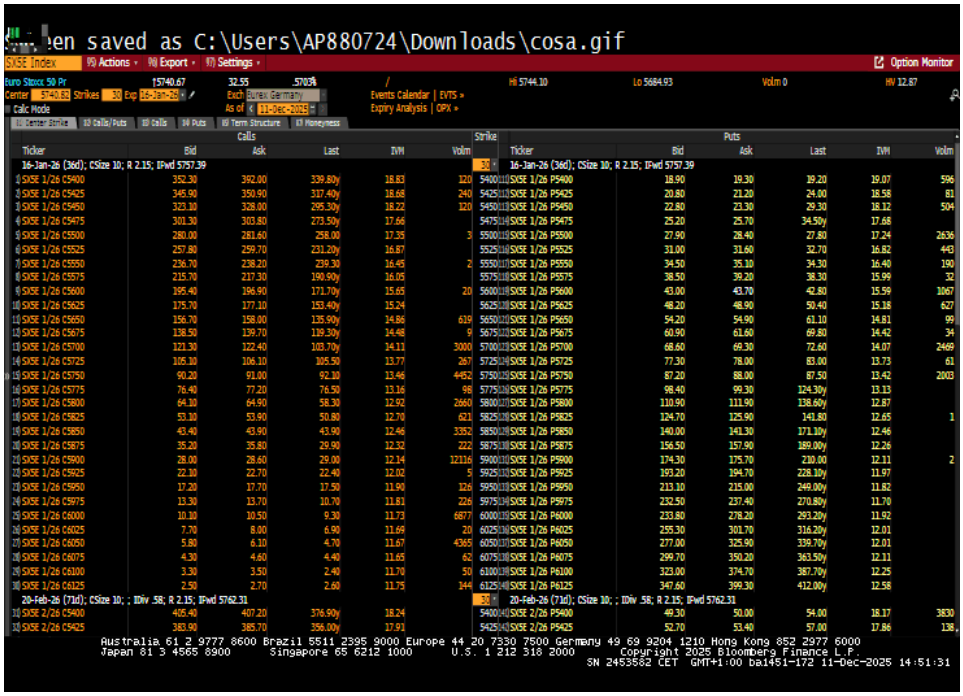

Figure 15: Bloomberg screenshot: Call and Put option prices with January 16, 2026 expiry.

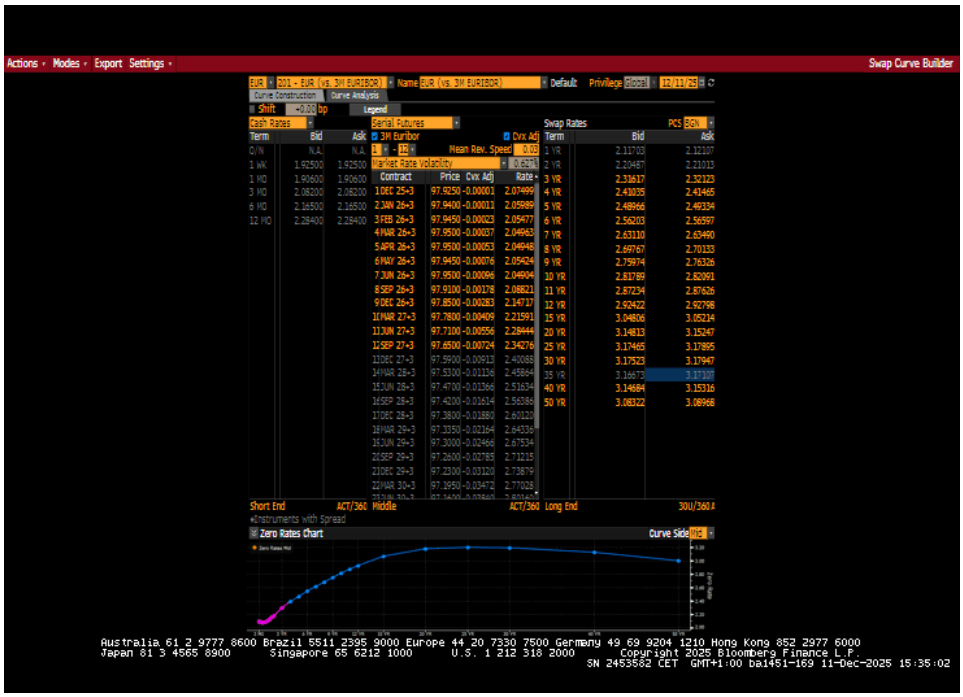

Figure 16: Bloomberg screenshot: 3-month Euribor.

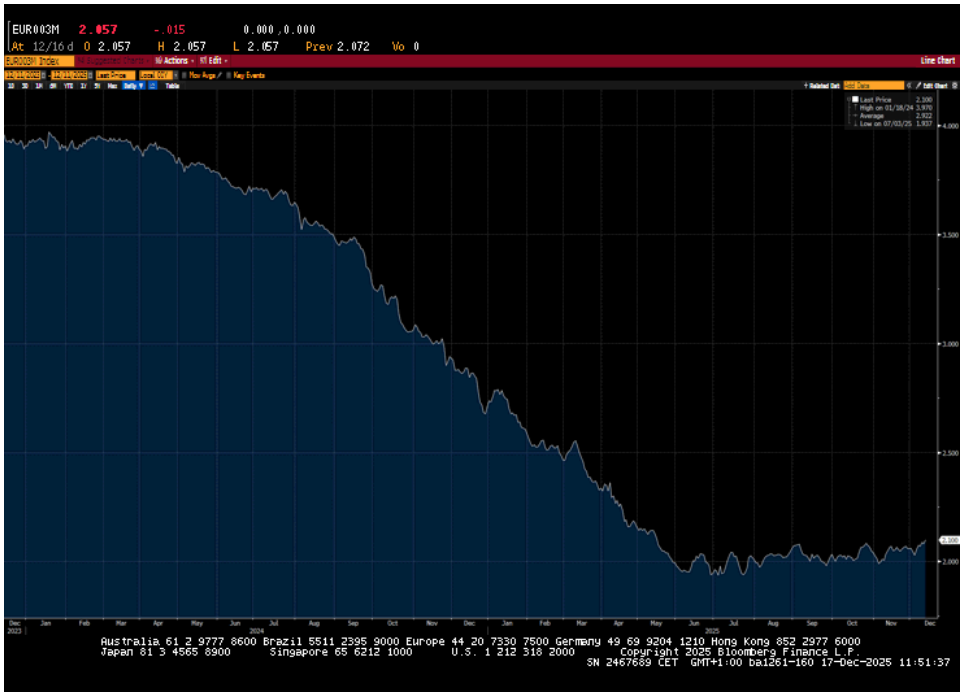

Figure 17: Bloomberg screenshot: 3-month Euribor historical series.

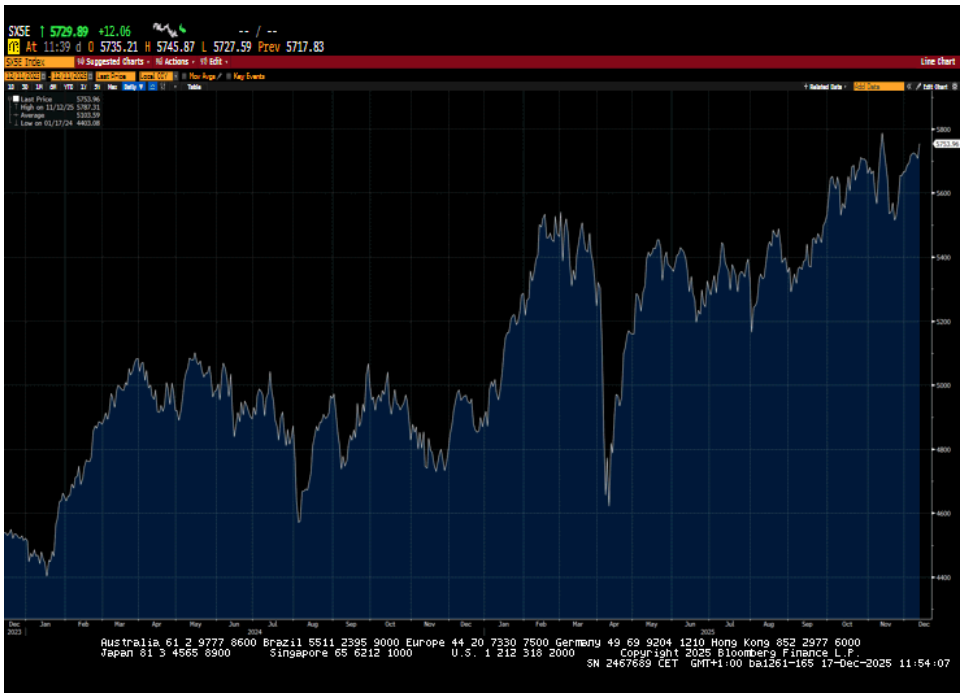

Figure 18: Bloomberg screenshot: EUROSTOXX50 historical series.

Figure 19: Bloomberg screenshot: EUROSTOXX50 Dividend Yield.

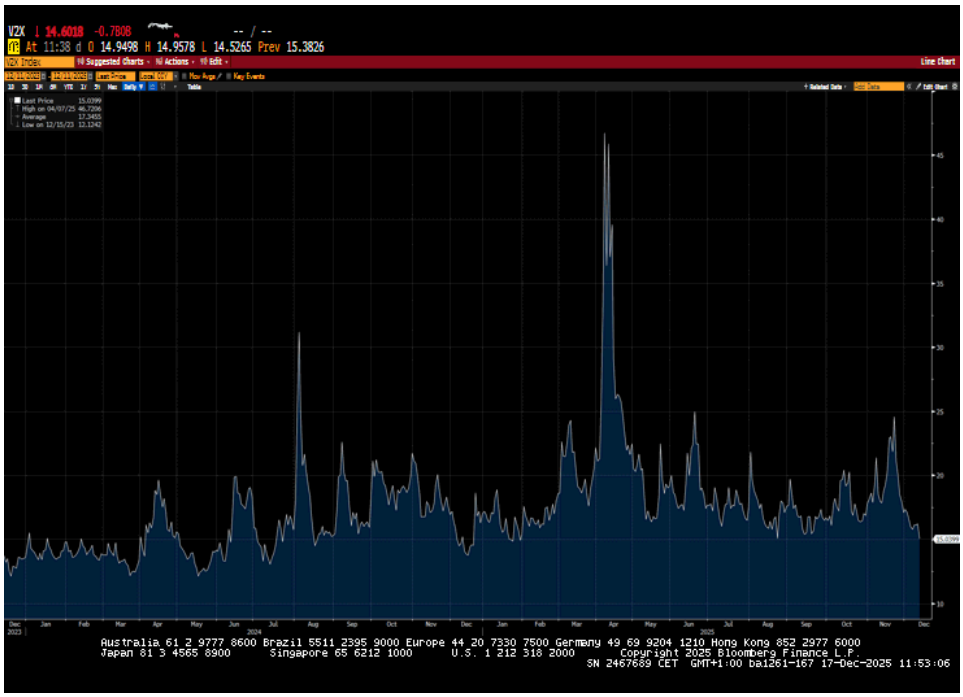

Figure 20: Bloomberg screenshot: VSTOXX historical series.

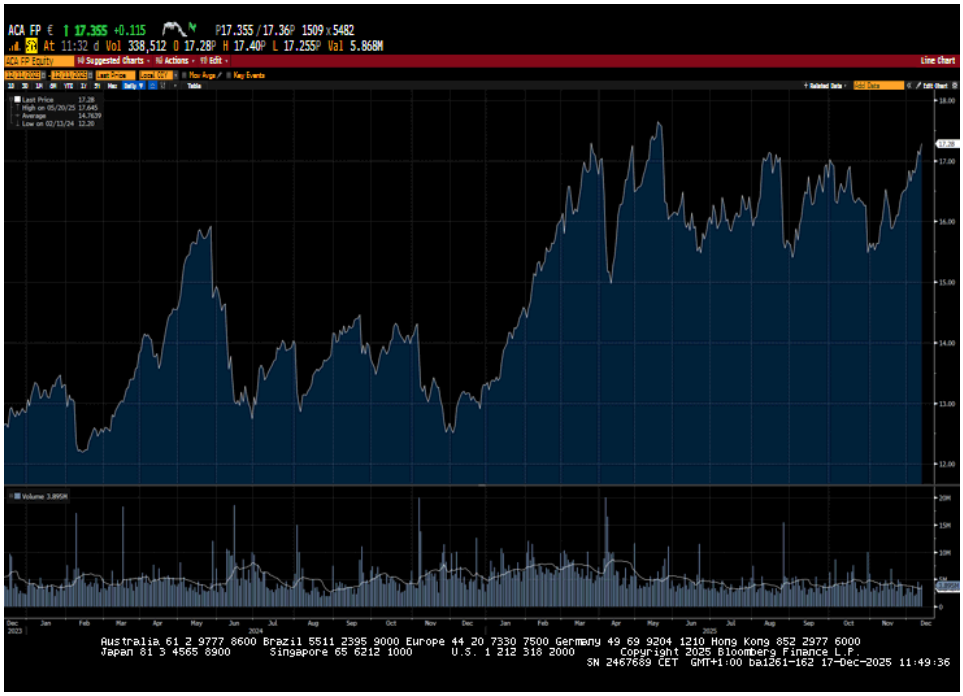

Figure 21: Bloomberg screenshot: Crédit Agricole price historical series.

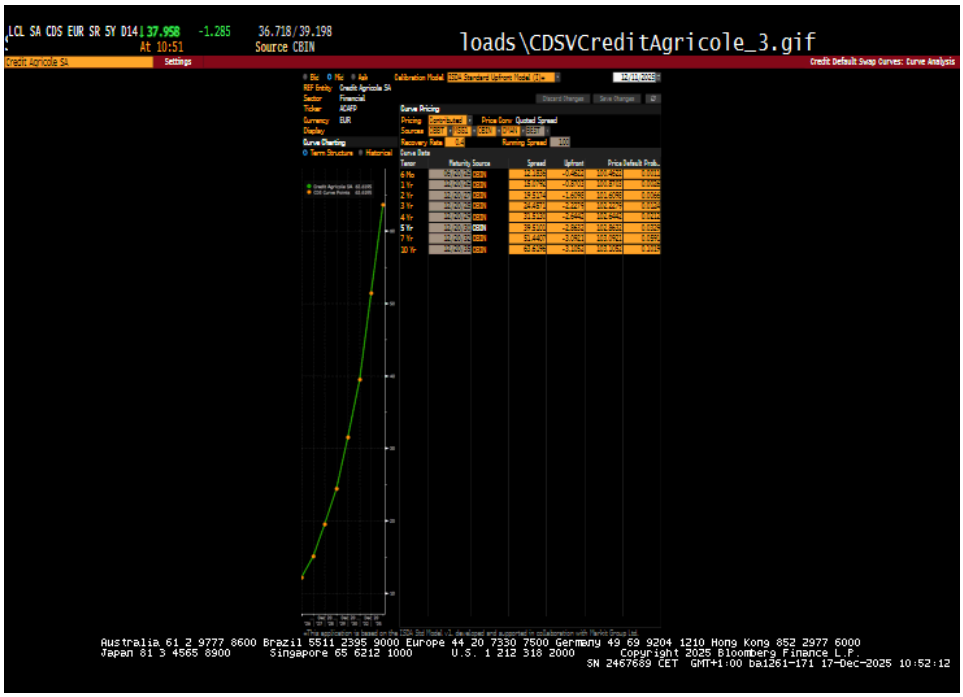

Figure 22: Bloomberg screenshot: Crédit Agricole CDS.